

- 7 (a) Neutron **A1**
 $x = 4$ **A1**
- (b) (i) The binding energy of a nucleus is the minimum energy required to completely separate the nucleus into its constituent nucleons (protons and neutrons) to infinity. **B1**

MC:	<i>Generally very poorly done. Most students were lacking in keywords in their definitions</i>
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(ii) $E_{\text{released}} = BE_{\text{products}} - BE_{\text{reactants}}$
 $= 139 (8.39) + (5 (8.74) - 235 (7.60))$ **C1**
 $= 210.51 \text{ MeV}$
 $= 3.36816 \times 10^{-11} \text{ J}$
 $= 3.37 \times 10^{-11} \text{ J}$ **A1**

MC:	<i>Generally not very well done. Most students missed out that the information given is the binding energy <u>per nucleon</u>, and hence did not multiply the mass numbers in their workings.</i> <i>A small handful had erroneous conversion of MeV into J.</i>
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(iii) $(\text{loss in mass}) c^2 = E_{\text{released}}$
 $(\text{loss in mass}) (3.00 \times 10^8)^2 = 3.36816 \times 10^{-11}$ **C1**
 $(\text{loss in mass}) = 3.74 \times 10^{-28} \text{ kg}$ **A1**

MC:	<i>Generally well done. Ecf was given should their answer in (ii) be incorrect.</i>
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- (iv) When more energy is being released in a nuclear reaction, more stable daughter nuclei are formed as a result. **B1**
 Since reaction 1 releases more energy than reaction 2, **B1**
 reaction 1 is more likely to occur. **B1**

MC:	<i>Generally well done. Ecf was given should their answer in (ii) be incorrect. Students' conclusion and explanations should be adjusted according to their answer in (b)(ii). (i.e. if the energy released obtained in (ii) is lower than reaction 2, students must conclude that reaction 2 is more likely to occur to attain credit)</i> <i>A small handful of students mixed up the concepts of spontaneity of reactions with the likelihood of reactions. Students are advised not to haphazardly introduce scientific terms outside of what the question has stated or is required to answer the question to avoid being marked down unnecessarily.</i>
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8(a)	Gravitational field is a region of space in which a mass will experience a non-contact gravitational force.	A1
CKW MC	- This definition was assessed in 2017 & 2018 H2 P3. - Refer to DHS “Gravitational Field” notes for reference. - Many candidates confused <i>gravitational field</i> with <i>gravitational field strength</i>. - An <i>area</i> is not the same as a <i>volume</i> or <i>region</i>. - For definitions, avoid using the root words again. Thus, candidates should explain the words <i>gravitational</i> and <i>field</i> separately.	
8(b)(i)	$\rho = \frac{M}{V} = \frac{M}{\left(\frac{4}{3}\pi r^3\right)} = \frac{0.73}{\left(\frac{4}{3}\pi(25 \times 10^{-3})^3\right)}$ $= 11000 \text{ kg m}^{-3}$	C1 A1
CKW MC	There was quite a number of students who did not use the correct formula for the volume of a sphere or know the importance of the word “diameter”.	
8(b)(ii)	Unit of τ = Unit of $\frac{2\pi^2 mL^2 \theta}{T^2}$ $\text{N m} = \text{kg m s}^{-2} \text{ m} = \frac{\text{kg m}^2}{\text{s}^2}$ Unit of θ Unit of $\theta = 1$ $\tau = \frac{2\pi^2 mL^2 \theta}{T^2}$ $1.2 \times 10^{-5} = \frac{2\pi^2(0.730)(1.80)^2 \theta}{(14.0 \times 60)^2}$ $\theta = 0.18136 \text{ rad}$ $= 0.18136 \left(\frac{360}{2\pi}\right)^\circ = 10(.4)^\circ$	C1 A1 A1
CKW MC	- Most students did not realise the unit of θ in the given equation is in radians. - Some students used $2m$ instead of m.	

8(b)(iii)	Radius of large sphere, $R = D / 2 = 15.0 \text{ cm}$ Radius of small sphere, $r' = d / 2 = 25 \text{ mm} = 2.5 \text{ cm}$ Thus, $r = r' + R + \text{air gap width}$ $= 15.0 + 2.5 + 0.1$ $= 17.6 \text{ cm}$	B1 C1 A0
CKW MC	- Quite a number of students attempted to use Newton's law of gravitation for this part using the given value of torque! - It is also not possible to use the given torque in (b)(ii) for this part since it is a different experiment, and the torque is different for $M = 158 \text{ kg}$ as can be shown that $LF_g = L \left(\frac{GMm}{r^2} \right) = 1.2 \times 10^{-5}$ $r = \sqrt{\frac{LGMm}{1.2 \times 10^{-5}}} = \sqrt{\frac{(1.8)(6.67 \times 10^{-11})(158)(0.73)}{1.2 \times 10^{-5}}} \neq 0.176 \text{ m}$	
8(b)(iv)	Using $s = (L / 2) \theta_1$, $4.1 \times 10^{-3} = (1.8 / 2) \theta_1$ $\theta_1 = 4.5 \times 10^{-3} \text{ rad}$ The uncertainty equation is given by $\frac{\Delta \theta_1}{\theta_1} = \frac{\Delta s}{s} + \frac{\Delta L}{L}$ $\frac{\Delta \theta_1}{4.5 \times 10^{-3}} = \frac{0.1}{4.1} + \frac{0.01}{1.80}$ $\Delta \theta_1 = 0.1 \times 10^{-3} \text{ rad}$ $\theta_1 = (4.6 \pm 0.1) \times 10^{-3} \text{ rad}$	A1 C1 A1
CKW MC	The max-min method will lead to the same answer.	

8(b)(v)	At θ_1, the torque caused by the gravitational attraction (between M and m) was equal to the opposing torque caused by the torsion in the wire. Thus, from (b)(ii), $LF_g = \frac{2\pi^2 mL^2 \theta}{T^2}$ $L \frac{GMm}{r^2} = \left(\frac{2\pi^2 mL^2}{T^2} \right) \theta_1$ $G = \frac{2\pi^2 L r^2 \theta_1}{MT^2}$ (Shown)	B1 C1 A0
	OR The maximum restoring force is equal to the gravitational force between the large and small spheres at a separation of r.. Thus, $ F_g = ma = m\omega^2 x_o$ $\frac{GMm}{r^2} = m \left(\frac{2\pi}{T} \right)^2 x_o$ $G = \left(\frac{4\pi^2}{T^2} \right) \frac{r^2}{M} x_o$ The amplitude x_o of the simple harmonic motion is approximately equal to $\frac{L}{2} \theta_1$. Thus, $G = \left(\frac{4\pi^2}{T^2} \right) \frac{r^2}{M} \left(\frac{L}{2} \theta_1 \right)$ $G = \frac{2\pi^2 L r^2 \theta_1}{MT^2}$ (Shown)	B1 C1 A0

CKW MC	Many candidates seem to have the misconception that gravitational force <i>is</i> the torque and/or the squared term for L gets omitted or incorrectly simplified in one step e.g. $\frac{GMm}{r^2} = \left(\frac{2\pi^2 mL^2}{T^2} \right) \theta_1$ $G = \frac{2\pi^2 L r^2 \theta_1}{MT^2}$	
8(c)(i)	When the rod rotates by an angle of θ_2, the laser beam deflects by an angle of $2\theta_2$ since the angle of incidence of the laser beam is equal to its angle of reflection. Thus, Using $s_2 = D_2(2\theta_2)$, $15.6 / 100 = (12.00)(2\theta_2)$ $\theta_2 = 0.00650 \text{ rad}$ $= 0.00650 \left(\frac{360}{2\pi} \right)^\circ$ $= 0.372^\circ$	C1 A0
	OR Using small angle approximation, $2\theta_2 \approx \frac{s_2}{D_2} \rightarrow \tan(2\theta_2) = \sin(2\theta_2) = \frac{s_2}{D_2}$ $\tan(2\theta_2) = \sin(2\theta_2) = \frac{15.6 / 100}{12.00}$ $\theta_2 = 0.372^\circ$	C1 A0
CKW MC	Candidates should show all relevant substitutions and conversion.	

8(c)(ii)	$\frac{3}{4}$ of a period is equal to 10 mins 22 s. Thus, $\frac{3}{4}T = ((10)(60) + 22) \text{ s}$ $T = \frac{2488}{3} \text{ s}$ From (b)(v), $G = \frac{2\pi^2 L r^2 \theta_2}{M T^2}$ $= \frac{2\pi^2 (1.80)(17.6/100)^2 (0.00650)}{(158)(2488/3)^2}$ $= 6.58 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ Unit of $G = \text{Unit of } \frac{2\pi^2 L r^2 \theta_2}{M T^2}$ $= \frac{\text{m m}^2}{\text{kg s}^2}$ $= \text{m}^3 \text{ kg}^{-1} \text{ s}^{-2}$	M1 C1 A1 A1
CKW MC	- Most students did not realise the unit of θ_2 is in radians, and conversion needs to be performed for the θ_2 in degrees in (c)(i) if used. - Radian is not an SI base unit.	
8(c)(iii)	T will be determined by taking the average of a large number of oscillations (about 60 instead of about 1 oscillation as in (b)(ii)) with the same actual uncertainty, thus the percentage (or fractional) uncertainty of T will be reduced significantly.	A1
CKW MC	Quite a number of students did not use appropriate and specific terms in their answers such percentage and/or actual uncertainty.	
8(c)(iv)	The rod was assumed to be light.	A1
CKW MC	- Most candidates did not provide the main reason for the difference in T leading to a difference in G. - The system rotates so slowly in 1 oscillation ($T = 14 \text{ min}$) that the damping of air resistance is negligible.	